Sludge Dryer



Wastely Aqua

WASTELY

Waste Water Reclamation

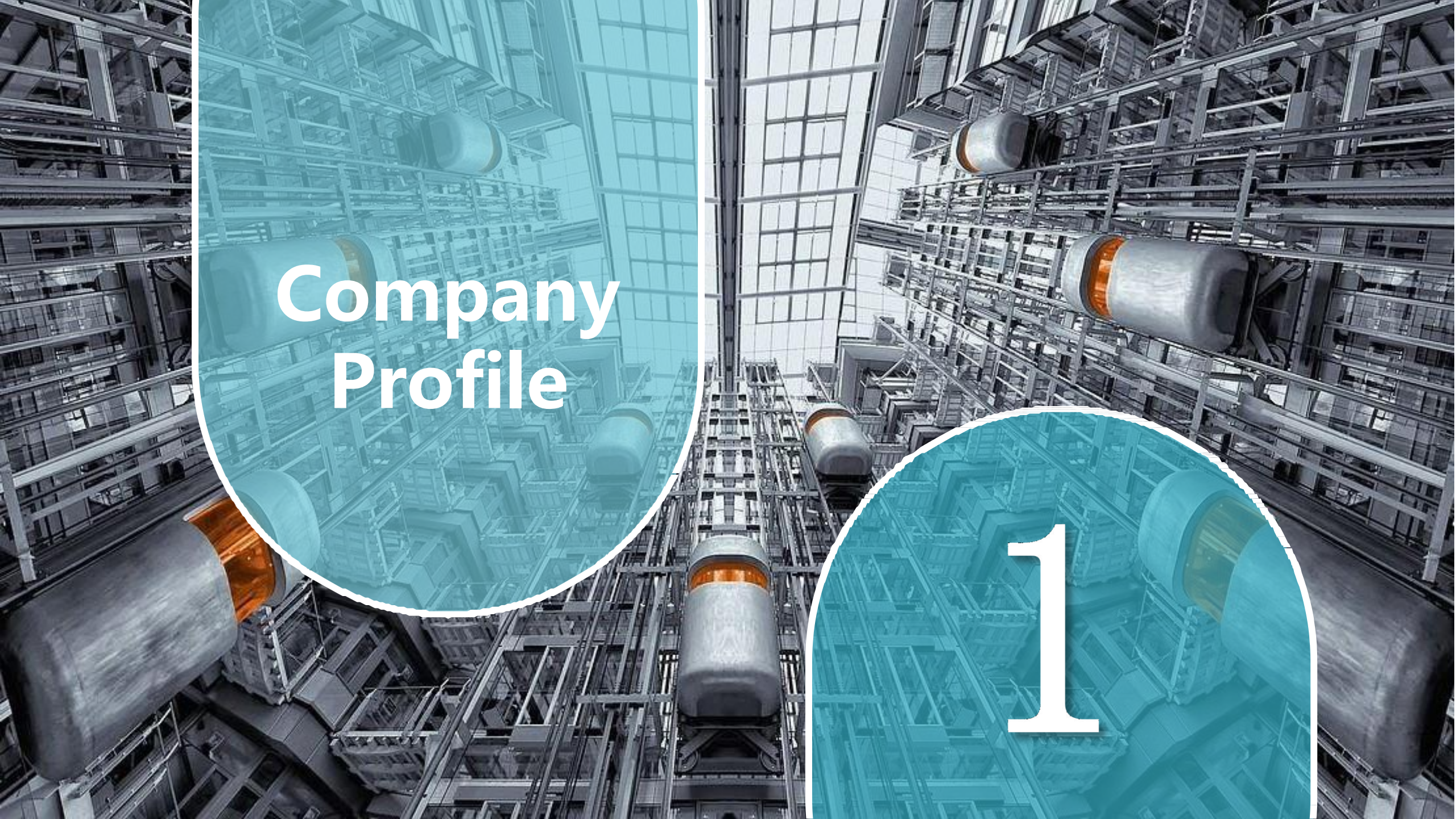
WASTELY
Waste Water Reclamation

About us 1

2 Sludge Dryer Introduction

Why Choose Us? 3

4 ECOPRO Sludge Dryer



Company Profile

1

Our Mission



Our Mission: To be your one-stop faithful partner of water and wastewater solution . Present leading equipments of environmental field.

Our Company



Design

According the project or clients requirement . Customized proposal for one-stop full line water and wastewater solution .



Fabrication

Water jet /sawing cutting
Sheet metal blanking
Frame welded
light gauge welding
Lathe machining
Polishing paint
Assembling



Inspection

Mainly including loading water test and hydraulic test for Sludge Dryer and idle equipments run test



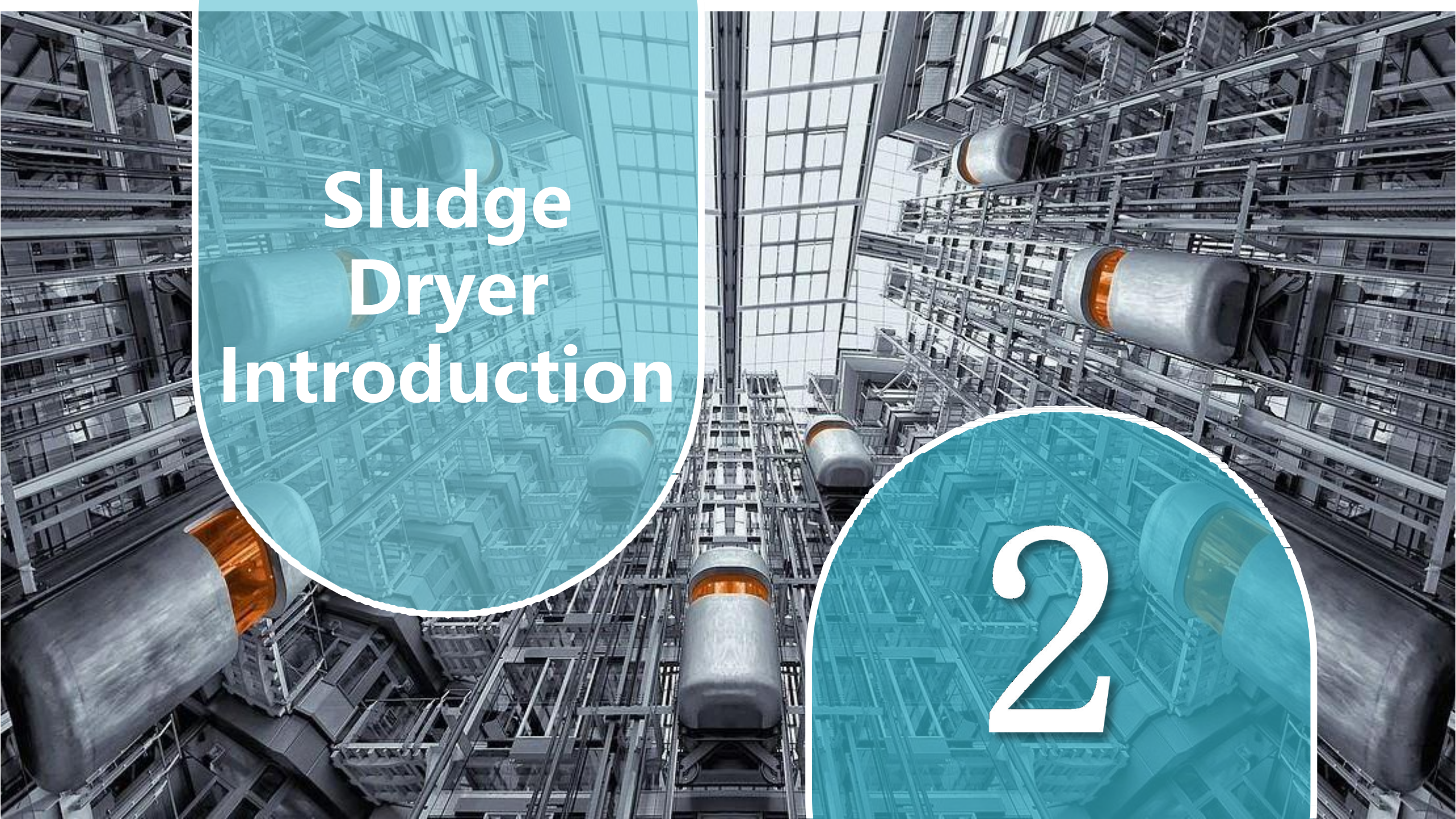
Packing&loading

According the equipments size .ECOPRO select suitable packing for unit . Wooden case for standard packing .



Shipping

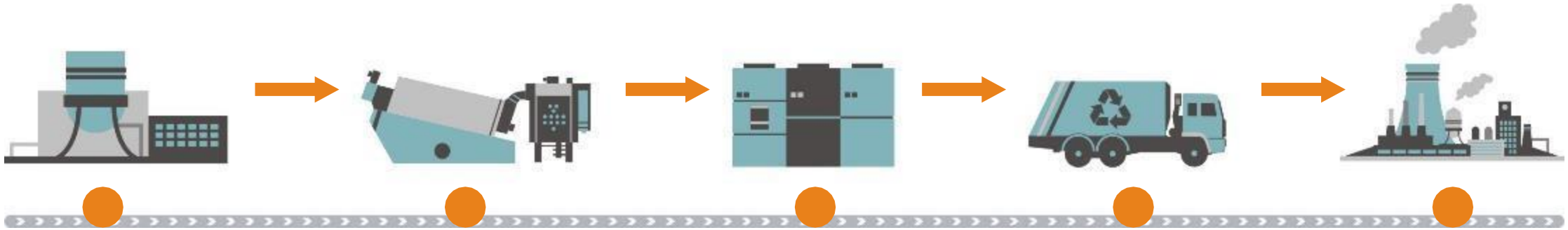
Sea transportation is the most recommand shipping terms .ECOPRO have long term cooperate forwarder company .Try best to send the units ASAP in reasonable cost .



Sludge Dryer Introduction

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Sludge Dryer Mechanism



Sewage Concentration Tank (sludge with 99% moisture)

Sludge dewatered to 83%-60% moisture

Sludge dried to 10%-50% moisture

Transport the dried sludge

Resource utilization

Sludge Dryer Mechanism

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Three types of main dryer, widely used in various industries

LOW TEMPERATURE WASTE HEAT SLUDGE DRYER
(Heat sourse Electricity+90℃ water)

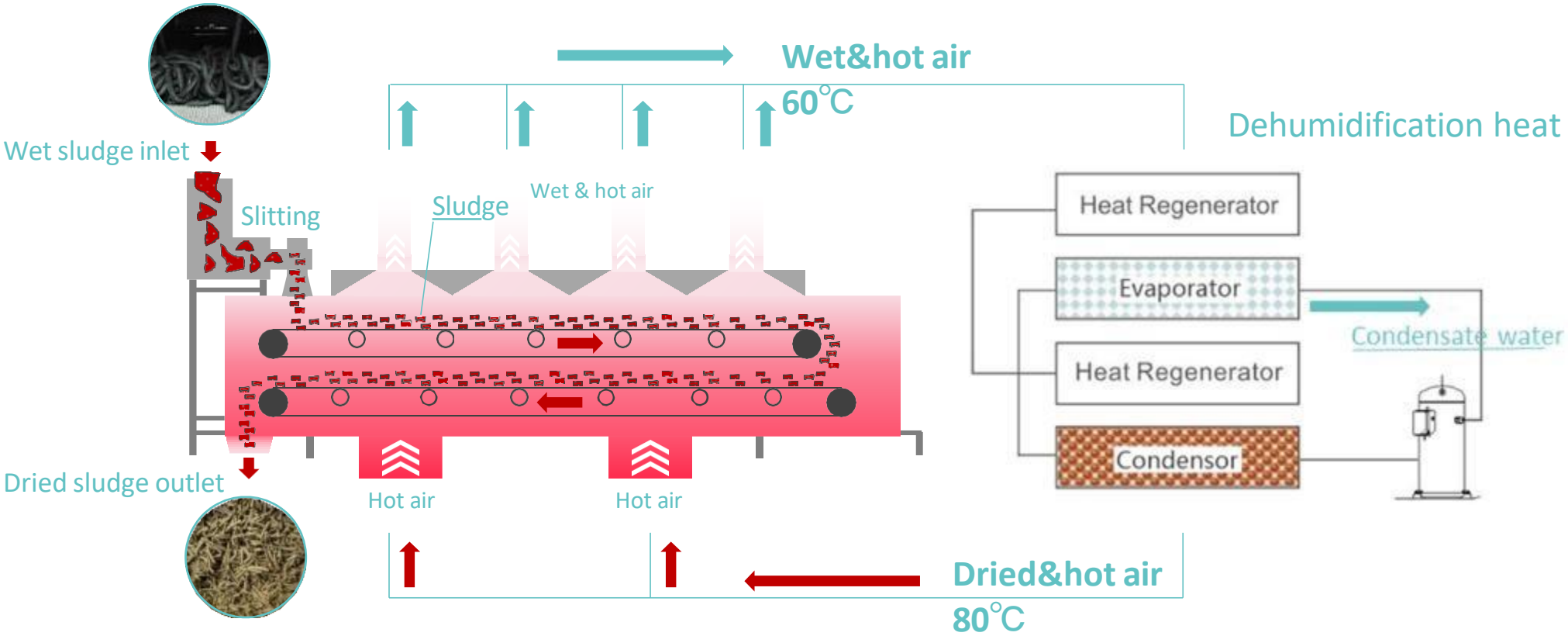
LOW TEMPERATURE BELT TYPE SLUDGE DRYER (Heat sourse Electricity)

LOW TEMPERATURE CONTAINER TYPE SLUDGE DRYER(Heat sourse Electricity)
Special for dryig the sludge cake outlet from Chamber Filter Press

Sludge Dryer Mechanism

Heat pump low temperature drying working principle

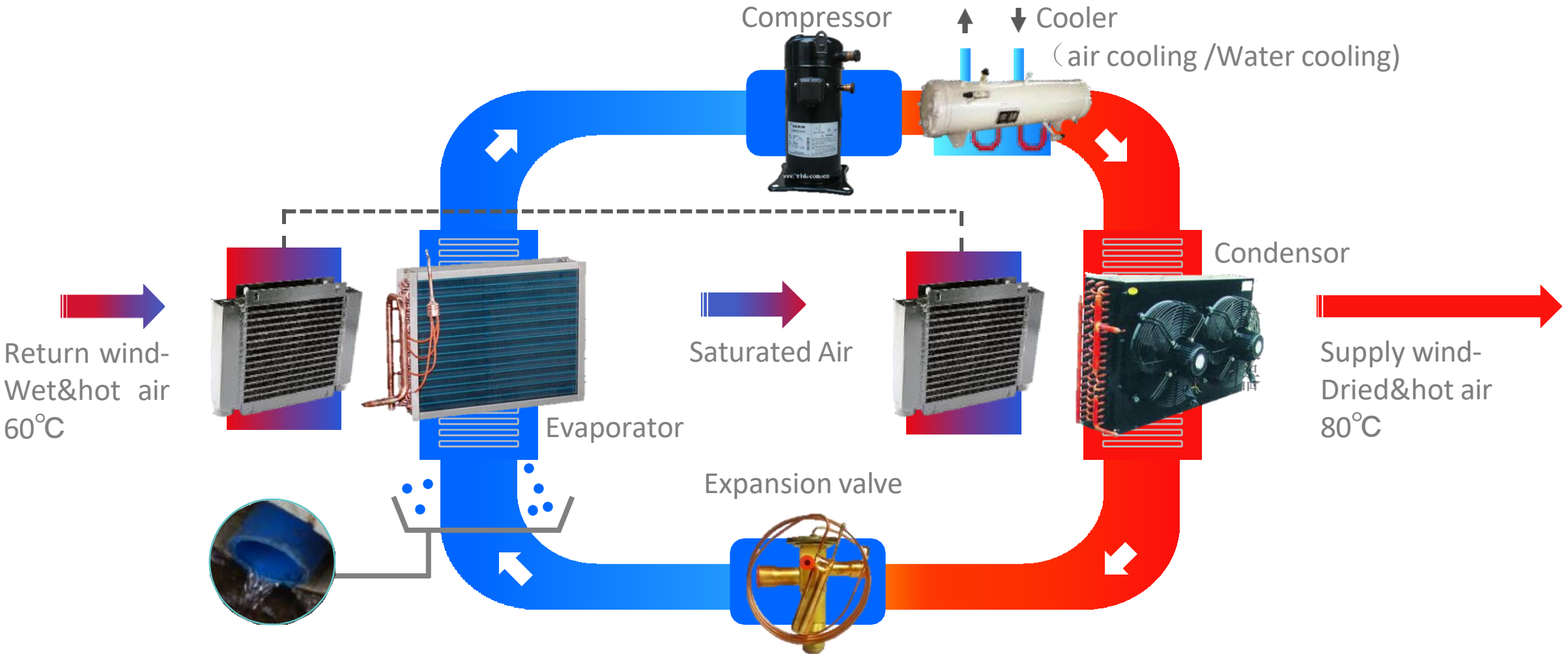
- Sludge dryer makes use of heat pump to carry out sludge through hot-air closed convectional recycling, condensing, dehumidifying and drying.
- Both drying medium and dehumidifying medium is hot air. Sludge absorbs heat from hot air to evaporate moisture. Moisture is carried out via hot air to condense. So there is no emission of waste heat during dehumidification drying.





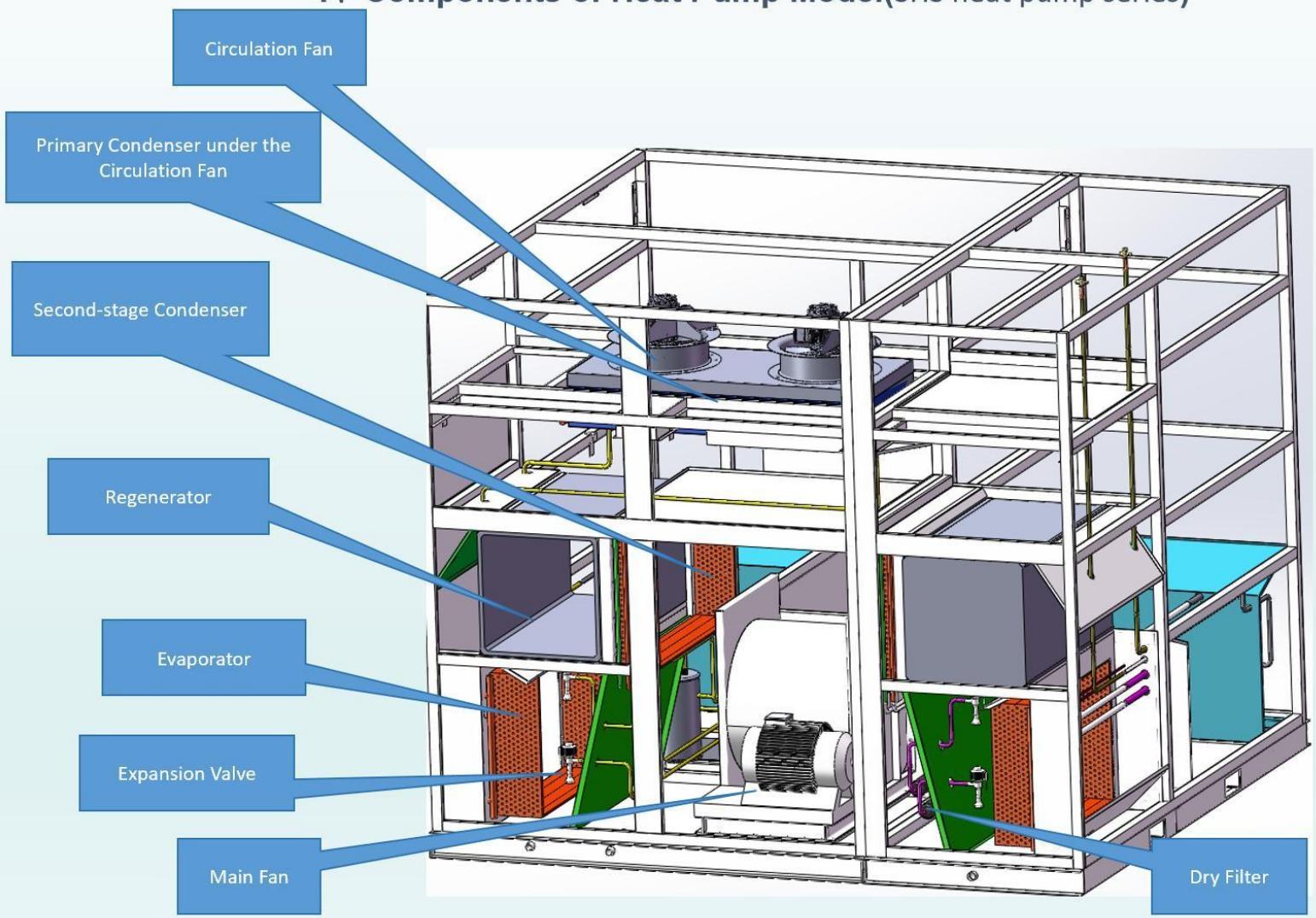
Sludge Dryer Mechanism

Heat Pump Principle	Dehumidification Heat Pump = Dehumidification (dehumidifying and drying) +Heat Pump (energy recycling) Heat source = water vapor phase change in the air to release condensation heat.



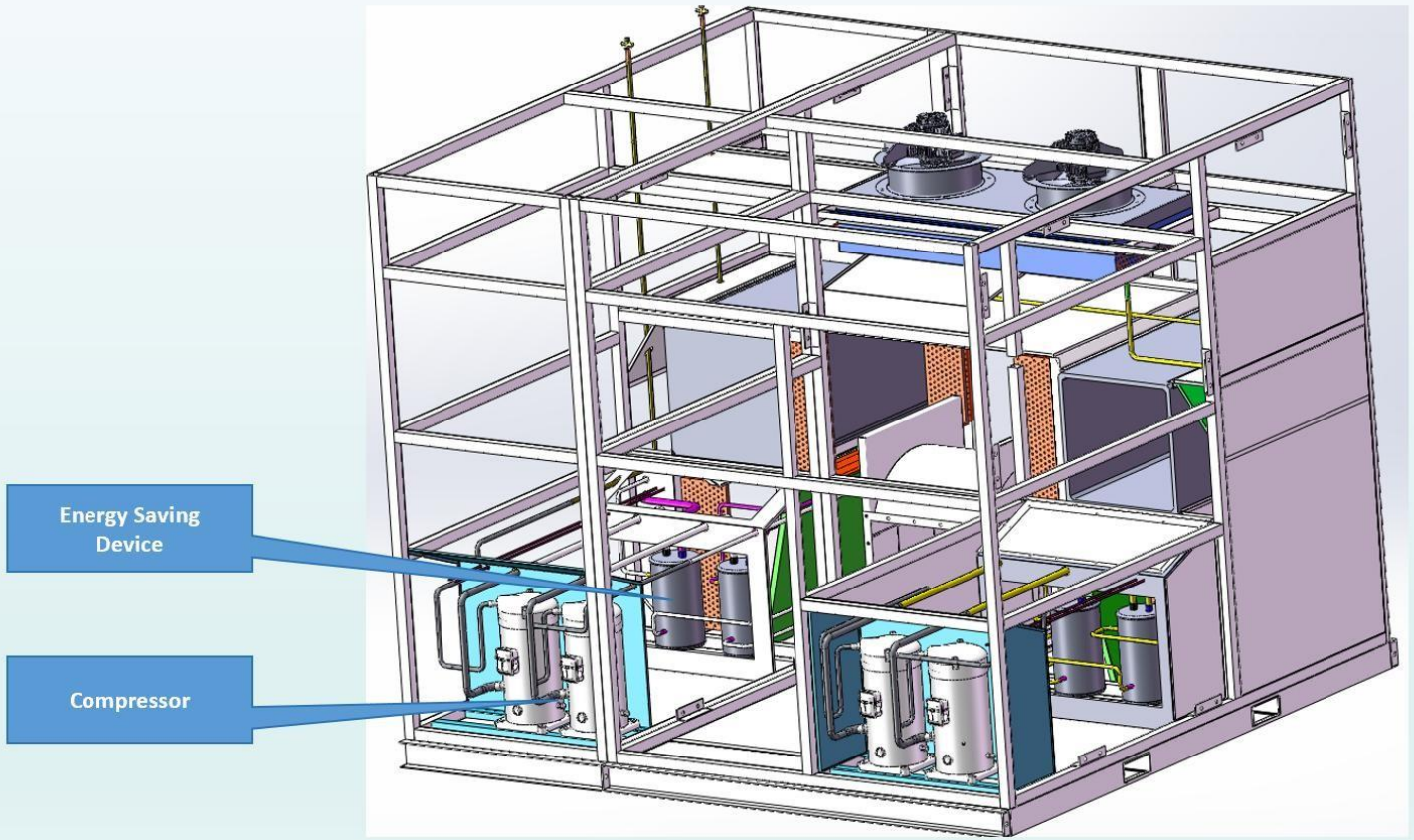
Sludge Dryer Mechanism

1、Components of Heat Pump Model(SHS heat pump series)



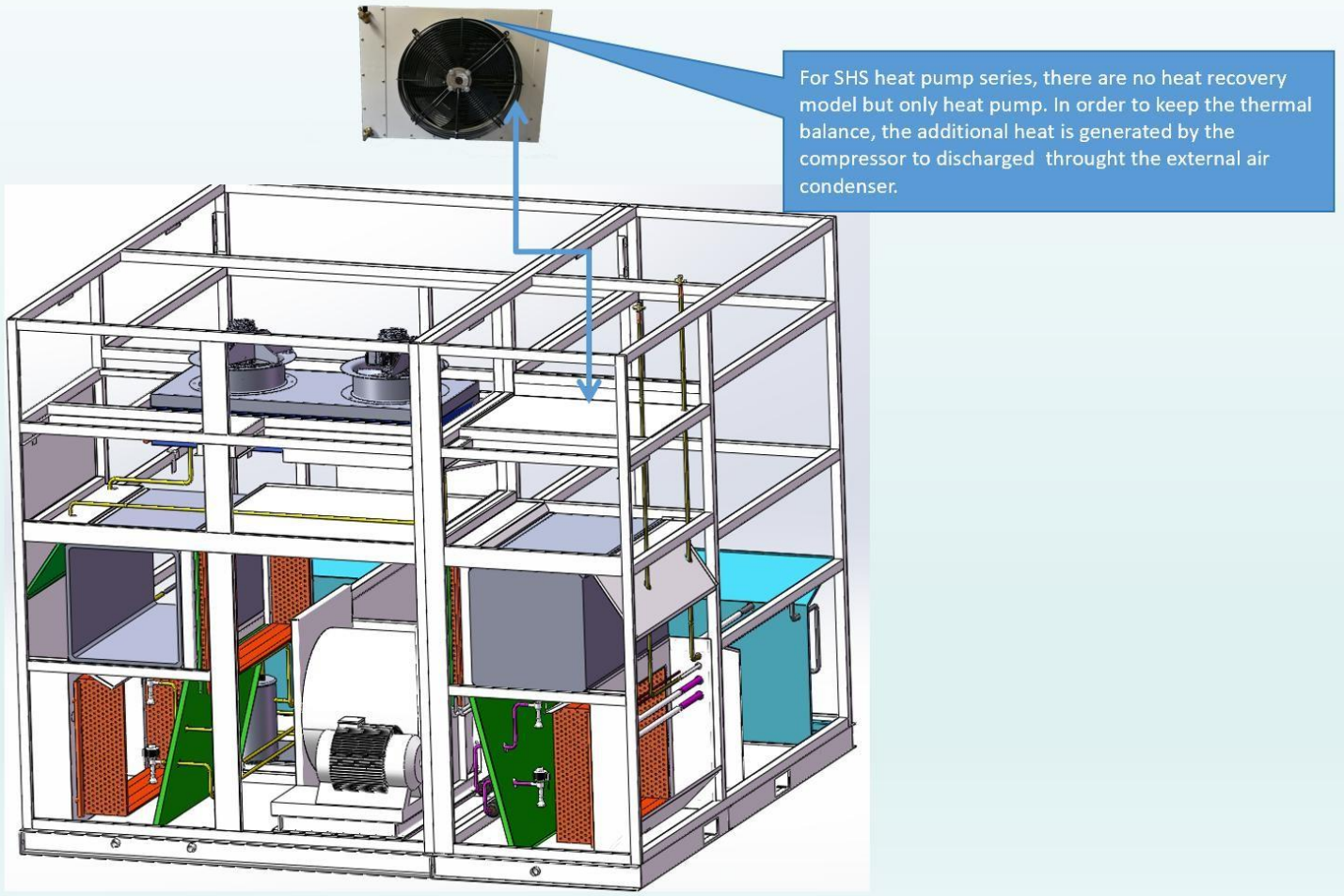
Sludge Dryer Mechanism

1、Components of Heat Pump Model(SHS heat pump series)



Sludge Dryer Mechanism

2、Instruction of Heat Pump Model(SHS heat pump series)

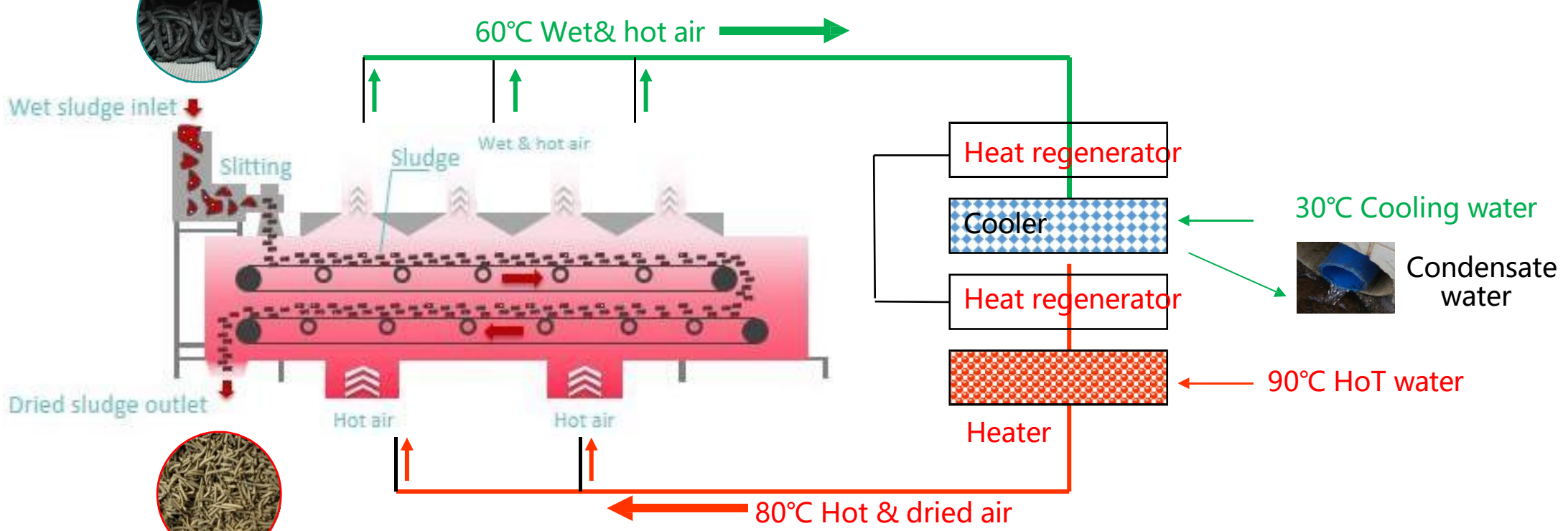


Sludge Dryer Mechanism

Low temperature waste heat dryer principle

- The waste heat dryer's heater and cooler used to condense and dehumidify the sludge by hot air circulation.
- Dehumidification recycle the latent heat of water vapor and sensible heat of air, without any waste heat discharge in the process.
- 90% of traditional sludge thermal drying system converts heat into exhaust heat loss (latent heat and hot air sensible heat)





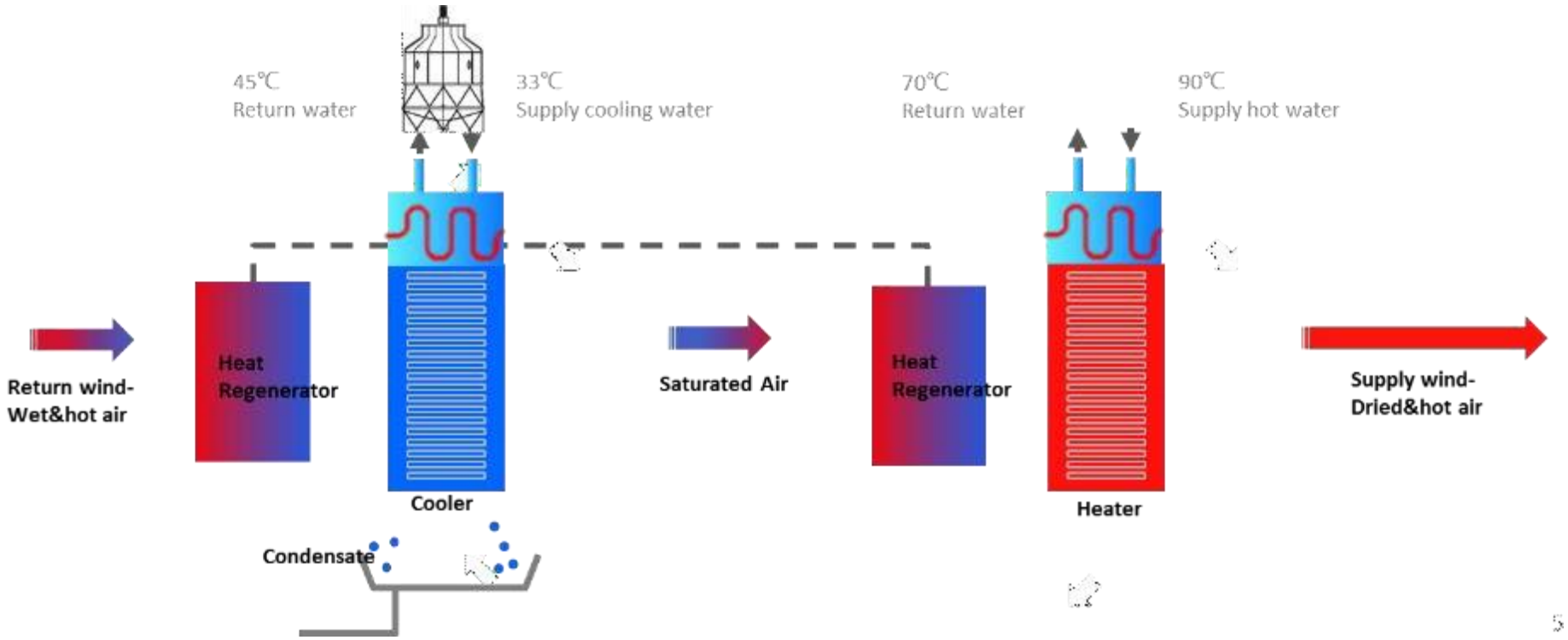


Sludge Dryer Mechanism

Waste-Heat-exchanger Dryer

Waste-Heat-Exchanger - Structure/Principle

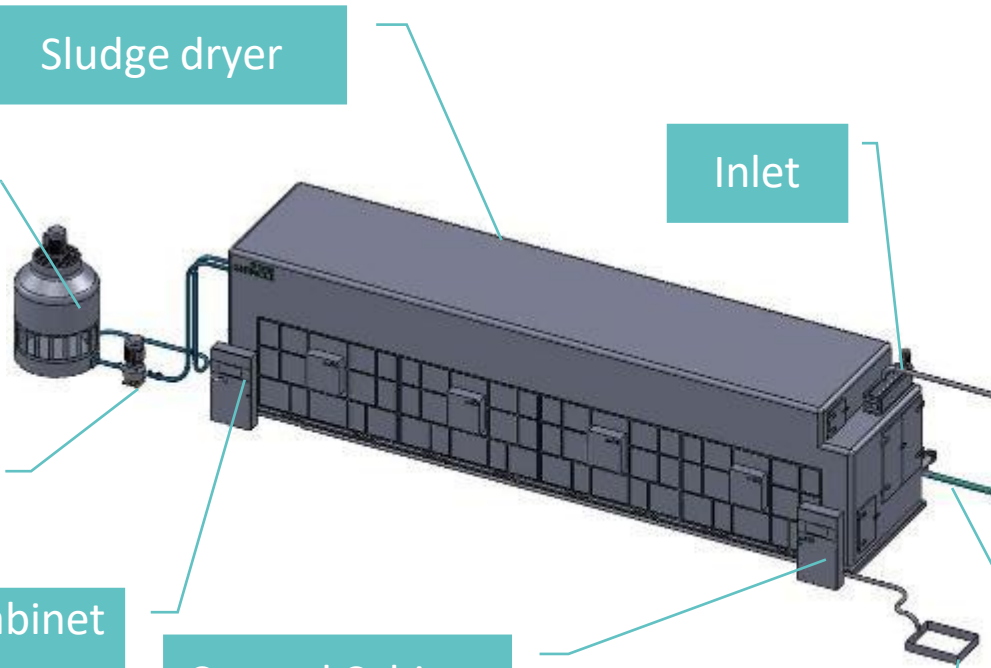
- No compressor (refrigeration system), only components of electric motor and fan with power consumption, it is only 1/4 that of Dehumidification Heat Pump Module.



- Cold water system makes the wet & hot air dehumidification and condensation. Inlet water 33℃, outlet water 45℃. According to the drying capacity of different modules, certain volume of hot water is to be provided.
- Hot water system heats up the dry air. Inlet water 90℃, outlet water 70℃. According to the drying capacity of different modules, certain volume of hot water is to be provided.

Sludge Dryer Mechanism

Drying system composition




Why
choose us
3

Why Choose Us ?

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1. Our stuff are all with *years of experience of wastewater treatment equipments* . And average work for *oversea business more than 5~8 years* . Especially on sludge dryer and sludge dehydrator .

2. Cooperate with the *biggest and most professional sludge dryer factory* in India . Know well local market and oversea market . Fabrication capacity reach over 260~300 *sets/year* .

3. Strictly according 5S stardard *inspection* the equipments during fabrication and loading. *Make record for test report of the idle running* before shippment .



Why Choose Us ?

One Stop Solution for Wastewater Treatment

Our core technology team is from leading sludge dryer, DAF factory , sludge dehydrator factory , filter press factory , sludge pump , polymer dosing pump factory and other well-known flag manufacturer . We have **huge resources integration ability**, and cooperate with many international well-known enterprise cooperation.In the future, Wastely Aqua will *continue to play the advantages of this kind of resource integration*, will be more integrated with different resources together, to **provide our partners with more different choices** .



Why Choose Us?

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Sludge Dehydrator

Sludge dehydrator is the best partner with sludge dryer. Where is a sludge dryer there is sludge dehydrator .



Rotary Screen



Rotary is very much popular used to separate the large particle of wastewater . Because of its superiority more and more widely adopted.

Resources Integration Ability

Lamella Clarifier

Lamella Clarifier is used for the prime stage to sedimentation the ss of the wastewater



Sludge Dryer

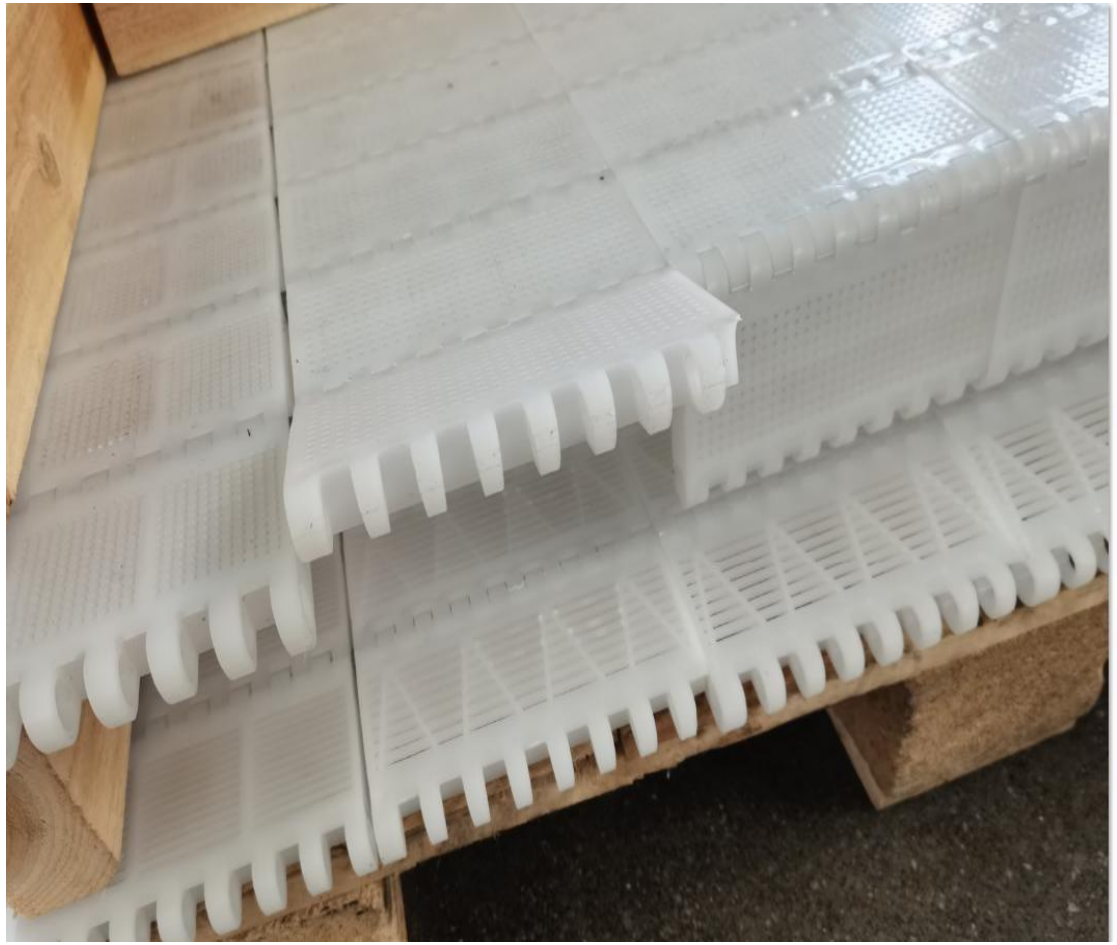


Sludge cake after sludge dehydrator may required more less moisture . Sludge dryer will be best choice .

Why Choose Us?

Newest PP mesh belt





Why Choose Us?

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Dust filter bag



304 stainless steel frame



Why Choose Us?

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Excellent Performance at worksite









ECOPRO Sludge Dryer

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ECOPRO Sludge Dryer

Technical Parameters

Model No.	Capacity (T/D)	Wet Sludge Moisture (%)	Dried Sludge Moisture (%)
SBDD2400FL	5	55	20
SBDD7200FL	10	70	20
SBDD9600FL	15	70	30
SBWHD20000	40	45	20
SBDD27000FSL	50	75	30
SBDD38400SL	60	65	30

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Startup
Frequency: Daily

Maintenance parts and operation instructions

1. Check if the hopper of slitter has dried sludge, and if so, remove it;
2. Check if sludge conveyance facilities has any dried sludge;
3. Check if cooling water system has any water;
4. After left unused for a long time, this product should be energized for over 12 hours before startup.
5. Confirm that all the equipment parts are intact in all the parts and the safety cover is under protection.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Dehumidification Heat Pump
Clean&Maintance
Frequency: Weekly

Maintenance parts and operation instructions

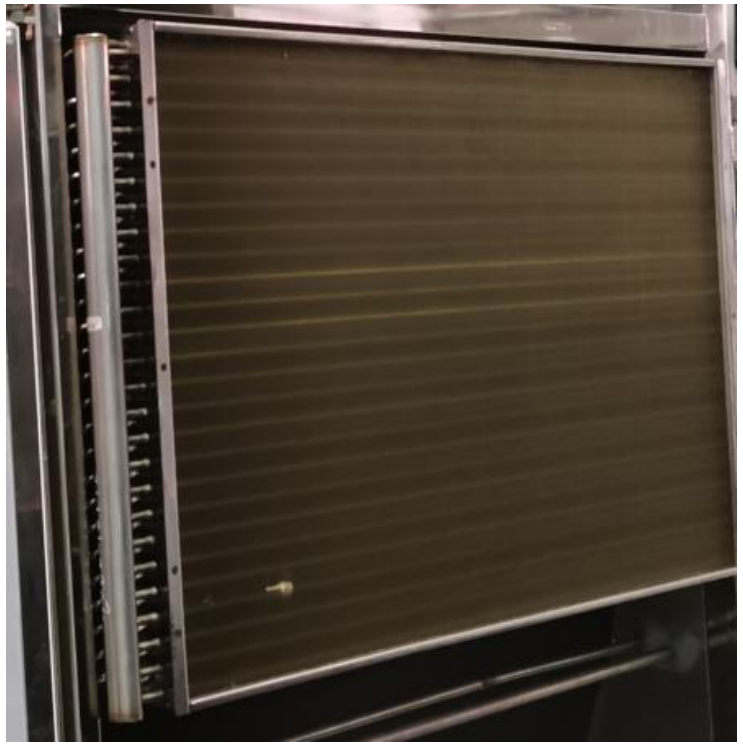
1. Fin Evaporator

1. Non-automatic cleaning device shall be used to clean evaporator at least once a week and ensure evaporator fin and copper pipe are clean.
2. Automatic cleaning device; examination shall be made at least once a month; if there is any residual attachment, user shall clean it with medicament.
3. Check it once a week. If it is not dirty or the dirt is not serious, it can be temporarily not cleaned.Under normal circumstances, clean the evaporator once a month (when cleaning, use a high-pressure water gun to flush the fins to prevent the fins from deforming) to ensure that the evaporator fins and copper tubes are clean.

1.4 Prevent the corrosion due to material sediment.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Dehumidification Heat Pump
Clean&Maintance
Frequency: Weekly

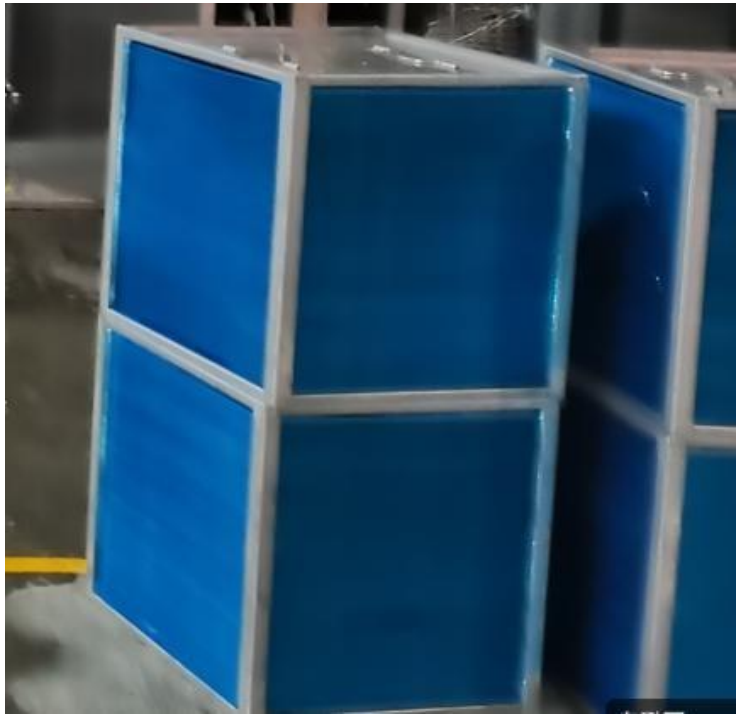
Maintenance parts and operation instructions

2. Fin Condenser

1. Check it once a week. If it is not dirty or the dirt is not serious, it can be temporarily not cleaned.Under normal circumstances, clean the evaporator once a month (when cleaning, use a high-pressure water gun to flush the fins to prevent the fins from deforming) to ensure that the condenser fins and copper tubes are clean.
2. In each season, user shall clean dust of and flush finned condenser at least one time.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Dehumidification Heat Pump
Clean&Maintance
Frequency: Weekly

Maintenance parts and operation instructions

3. Heat Regenerator, Pump Cavity, Water Tray

1. Clean and flush the regenerator, heat pump inner cavity and water receiving tray every week to ensure that the air duct is smooth and improve the drying efficiency of the equipment.
2. In each season, user shall clean dust of and flush heat regenerator at least one time.
3. In case of serious pollution, user should increase the times of dust removal and flushing.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Dehumidification Heat Pump Clean&Maintance Frequency: Weekly & Monthly

Maintenance parts and operation instructions

4. Filter (plate filter, bag filter)

1. The operator should clean the filter screen with compressed air no less than once a week (when blowing the plate filter, it needs to be blown from the reverse side, so as to prevent the dust from being pressed into the filter by the pressure of the front blow).
2. The warranty staff cleans the filter once a month. The filter should be replaced immediately if the filter is damaged or the ash accumulation is heavy.
3. The fixed replacement cycle of filter is 90 days to 180 days. Replacement cycle shall be adjusted according to applications.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Slitter and Inlet Hopper
Frequency: Weekly & Half-yearly

Maintenance parts and operation instructions

1. Noodle Comb

1. If there is abnormal noise in the face comb, it should be shut down for inspection immediately. Check the copper comb every half a month. If the following materials are uneven, you can first remove the hopper mud to observe whether the gap between the two sides of the round knife is balanced. If the gap is normal, remove the copper comb to clean it. an examination.
2. According to the characteristics of the sludge, the replacement cycle of the face comb is 6-12 months. (Replacement materials and costs are calculated separately)

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Slitter and Inlet Hopper
Frequency: Monthly

Maintenance parts and operation instructions

2. Bearing

1. Check the bearing temperature every two weeks, usually 10℃ to 40℃ higher than ambient temperature.
2. Check the bearing sound with stethoscope every month.
3. Lubricating grease shall be replaced every year.
4. Bearing shall be replaced every 1 year to 3 years.
5. Replacement materials and other costs are calculated separately.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Slitter and Inlet Hopper
Frequency: Weekly

Maintenance parts and operation instructions

3. Gear wheel

1. Check the gear wheel weekly to see if teeth are broken, worn or has pitting; remove sundries and dust.
2. Add industrial gear oil weekly.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Slitter and Inlet Hopper
Frequency: Monthly

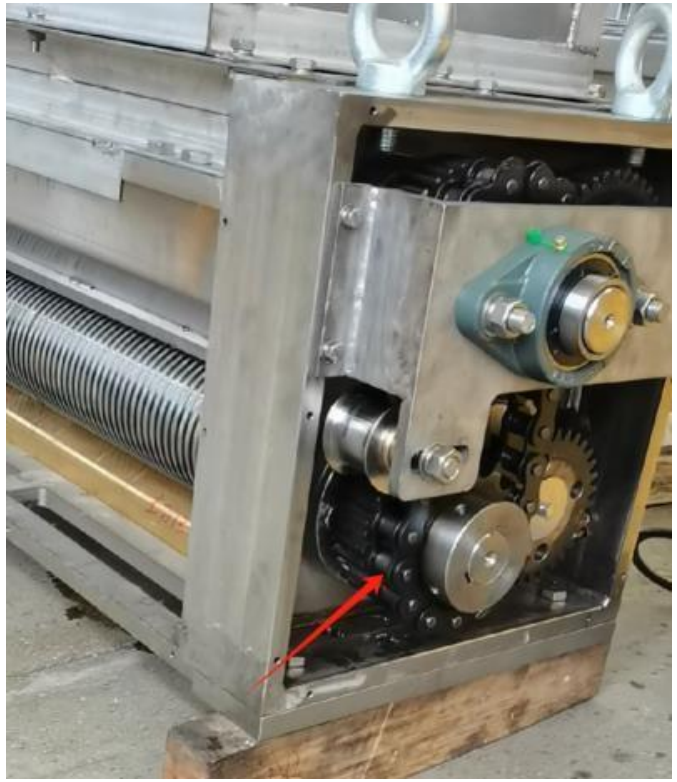
Maintenance parts and operation instructions

4. Rotary level switch

Every week, check if the blade of the rotary paddle switch has any sludge cake or lump, cannot rotate or the blade falls off.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Slitter and Inlet Hopper
Frequency: Monthly

Maintenance parts and operation instructions

5. Chain wheel

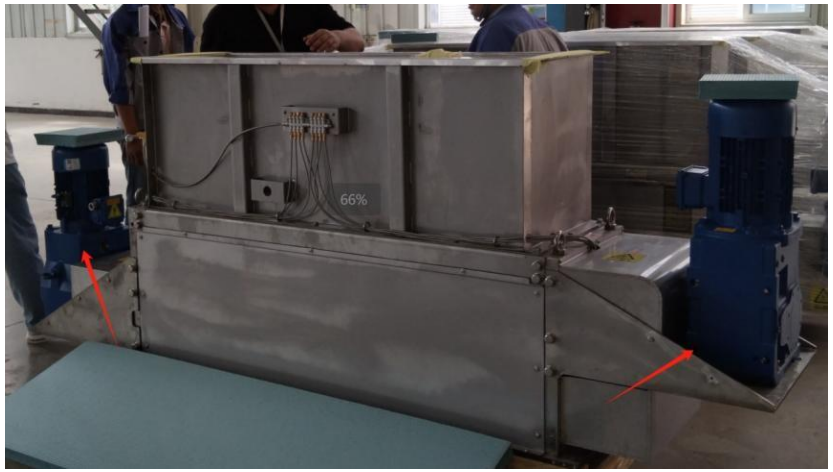
Every week, check the chain wheel to see if it is worn or if there are broken teeth or shift; butter shall be added weekly.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program

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Maintenance parts and operation instructions



Slitter and Inlet Hopper
Frequency: Monthly & Half-Yearly

6.Geared motor

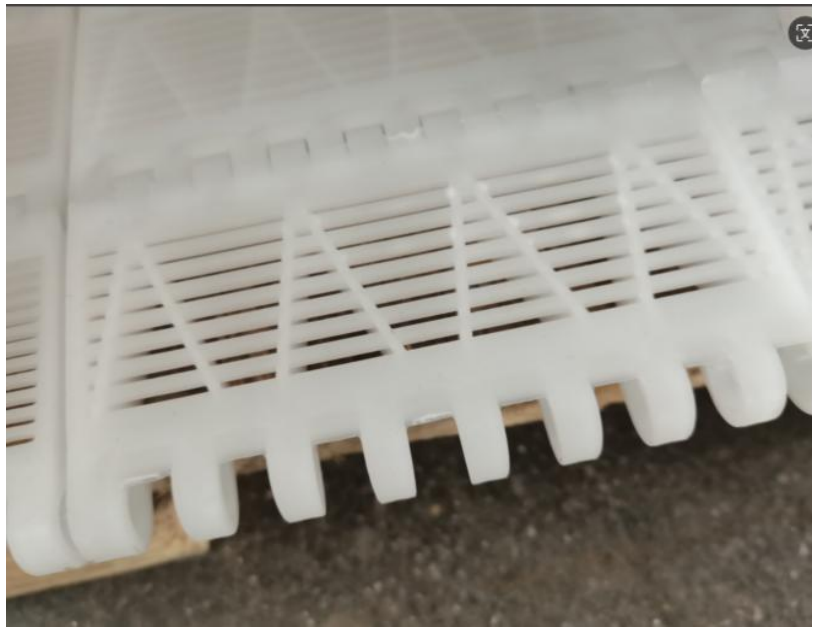
Check the reducer once a month for abnormal noise and vibration, oil leakage, oil temperature;

Check whether the operating voltage and current are normal once a month;

Check the oil quantity of the reducer lubricating oil (oil window 2/3) and whether it is carbonized or blackened every quarter. The carbonized and lubricated oil needs to be replaced with new lubricating oil.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Upper Belt, Lower Belt
Frequency: Monthly

Maintenance parts and operation instructions

1. PP Mesh Belt

Check the usage of the PP mesh belt once a month for deformation, cracking, or damage.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Upper Belt, Lower Belt
Frequency: Monthly

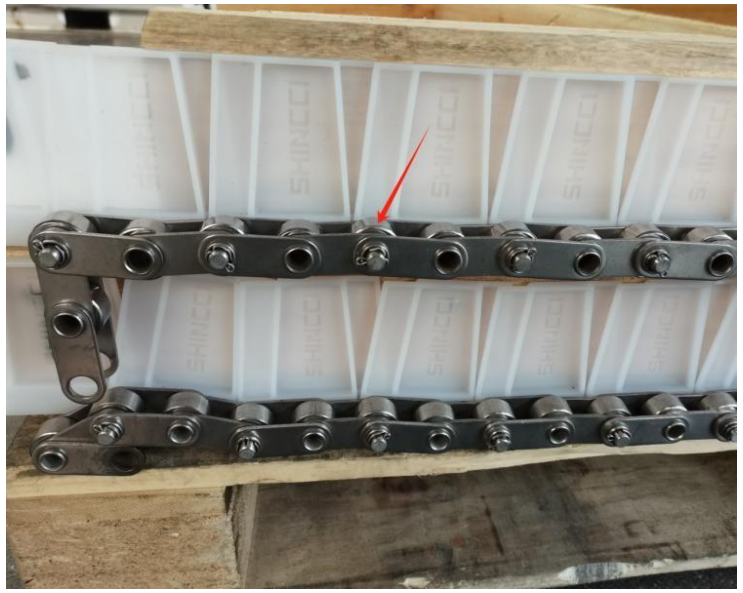
Maintenance parts and operation instructions

2. Bearing

1. Every month, check the temperature of the bearing; generally, bearing temperature shall be 10℃ to 40℃ greater than ambient temperature.
2. Every month, check the sound of the bearing with stethoscope.
3. Lubricating grease should be replaced yearly.
4. The bearing’s replacement cycle is 1 year to 3 years.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Upper Belt, Lower Belt
Frequency: Monthly

Maintenance parts and operation instructions

3. Chain wheel

Every week, check the chain wheel to see if it is worn or if there are broken teeth or shift; butter shall be added monthly.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Upper Belt, Lower Belt
Frequency: Monthly & Half-Yearly

Maintenance parts and operation instructions

4. Speed reducer

1. Check the speed reducer for possible abnormal sound, vibration, oil leakage and oil temperature, every month.
2. Replace the lubricating oil every halft year.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Shaftless Screw Conveyor
Frequency: Monthly

Maintenance parts and operation instructions

1. Shaftless spiral

1. Clean the inside of the shaftless spiral cavity once a month;
2. Check once a month if there is no abnormal vibration and abnormal sound during the shaftless spiral operation.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Shaftless Screw Conveyor
Frequency: Monthly

Maintenance parts and operation instructions

2. Bearing

Check the bearing temperature and add butter once a month; bearing replacement cycle is 2-3 years.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Upper Belt, Lower Belt
Frequency: Monthly & Half-Yearly

Maintenance parts and operation instructions

3. Worm Reducer

1. Check the reducer once a month for abnormal noise and vibration, oil leakage, oil temperature;
2. Check whether the operating voltage and current are normal once a month;
3. Check the oil quantity of the reducer lubricating oil (oil window 2/3) and whether it is carbonized or blackened every quarter. The carbonized and lubricated oil needs to be replaced with new lubricating oil.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program

Maintenance parts and operation instructions



Electrical system and heat pump accessories

1、Electric Leakage protection Switch (Monthly)

Check leakage circuit breakers monthly to see if they can work in an reliable way. Only when leakage circuit breakers can work normally after having been tested for three times can them be used.

2. Overload Protector (Half-yearly, yearly)

1. Check the rated current value of the thermal component of the thermal relay every six months to see whether it is equivalent to the rated current value of the motor;
2. Power-on verification once a year.

3. Contactor (Quarterly)

1. Quarterly, check if the contactor's movable and stationary contacts are aligned and if three phases are closed.
2. Quarterly, check auxiliary contacts to see if they act flexibly.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Electrical system and heat pump accessories

Maintenance parts and operation instructions

4. Electric Box (Monthly)

1. Check the electrical box once a month for water leakage, whether the cooling fan and shutters are well ventilated, and ensure that the interior is dry and ventilated;
2. Check quarterly whether the internal connection terminals of the electric box are firm.

5. Frequency Converter (Quarterly)
Dedust the inverter every quarter (depending on the amount of dust on site).

6. Programmable logic controller (PLC) (Monthly)

1. Check and confirm the working temperature of PLC is below 60℃ once every week.
2. Check whether wiring is reliable once every quarter.
3. Check whether wiring is reliable quarterly.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Electrical system and heat pump accessories
Frequency: Daily & Monthly

Maintenance parts and operation instructions

7. Touch screen

Before startup every time, user shall clean the touch screen with a piece of dry cloth, and clean dirty fingerprint and oil stain with glass cleaner.

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program





Electrical system and heat pump accessories
Frequency: Monthly

Maintenance parts and operation instructions

8. Main fan, circulating fan

1. Check the current, voltage and resistance of the main fan and circulating fan no less than once a month;

8.2 Every two weeks, the main fan needs to be cleaned of mud powder to prevent the high pressure alarm of the equipment caused by the clogged air duct;

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program



Electrical system and heat pump accessories
Frequency: Monthly

Maintenance parts and operation instructions

9. Compressor

Check the current, voltage and resistance of the compressor no less than once a month;

ECOPRO Sludge Dryer

Heat Pump Sludge Dryer Maintenance Program

Equipment operation observation Frequency: Monthly	Maintenance parts and operation instructions After the maintenance is completed every month, turn on the equipment and check whether the operating conditions and sludge drying effect are normal.

ECOPRO Sludge Dryer

Grease And Oil Specification

Part Name	Grease & Oil Model	Refueling Cycle	Operating Temperature	Viscosity
Bearing (Mesh Belt)	FM222	Once per three months	（-40℃~180℃）	ASTM D 445
Bearing (Blower)	RAREMAX SUPRE N	Once a month	（-40℃~200℃）	2#~3#
Bearing (Slitter & Bridge-breaker)	Great Wall General Grease 00# (Boutique Semifluid)	Once per half month (Automatic Refueling）	（-40℃~180℃）	40℃/00#
Motor Gearbox (Mesh Belt)	Shell Tivela Oil S 680	Once a year (1st time: 3months)	（-40℃~180℃）	40℃/680
Motor Gearbox (Slitter)	Shell Omala Oil 220	Once a year (1st time: 3months)	（-40℃~180℃）	40℃/220
Bearing (Screw Conveyor)	Great Wall General Grease 2#	Once a month	（-40℃~180℃）	40℃/2#
Bearing (Bridge-breaker of Silo)	Great Wall General Grease 2#	Once a month	（-40℃~180℃）	40℃/2#
Motor Gearbox (Screw Conveyor)	Shell Omala Oil 220	Once a year (1st time: 3months)	（-40℃~180℃）	40℃/220
Motor Gearbox (Bridge-breaker of Silo)	Shell Omala Oil 220	Once a year (1st time: 3months)	（-40℃~180℃）	40℃/220